(10) Publication No: **IN202621028942 A1**
(43) Publication Date: 01-05-2026
(22) Date of Filing: 11-03-2026
Journal No: 18/2026

(19) Intellectual Property Office, India

(12) INDIAN PATENT APPLICATION

(54) Title: **A method for operating a touchless mouse using gesture recognition to enhance user interaction and improve accessibility**

(51) International Classification:
G06F 3/01, G06N 20/00, G06F 3/0346, G06F 3/033 G06F 3/038

(21) Application No: **202621028942**

(31) Priority Document No: --

(32) Priority Date: --

(86) International Application No: --
Filing Date: --

(87) International Publication No: --

(61) Patent of Addition to Application No: --
Filing Date: --

(62) Divisional to Application No: --
Filing Date: --

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(57) Abstract:

This invention relates to a method for operating a touchless mouse using finger gesture recognition to enhance user interaction and improve accessibility. The system comprises a wearable finger band equipped with a high-resolution motion sensor that captures hand movements and gestures in three-dimensional space, translating them into cursor movements and commands on a digital interface. An AI-driven recognition algorithm processes sensor data, generating control commands for various digital systems, including IoT devices and medical equipment. This dynamic and adaptive algorithm leverages machine learning to tailor the user experience based on individual interaction patterns. Designed ergonomically for comfort, the finger band facilitates intuitive interaction, particularly benefiting individuals with mobility impairments. By eliminating the need for physical contact with traditional input devices, this invention promotes a more inclusive and user-friendly digital experience across diverse environments, from home to professional settings, thereby enhancing overall accessibility in technology.

FORM 2

THE PATENT ACT 1970

(39 OF 1970)

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THE PATENTS RULES, 2003

COMPLETE

SPECIFICATION

(See section 10 and rule 13)

TITLE OF THE INVENTION:

"A Method for Operating a Touchless Mouse Using Gesture Recognition to Enhance User Interaction and Improve Accessibility"

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PREAMBLE TO THE DESCRIPTION:

The following specification particularly describes the invention and the way it is performed.

DESCRIPTION OF THE INVENTION:

Technical Field of the Invention

[0001] The present invention relates to AI-driven, machine learning, and assistive technology. More specifically, the invention relates to a touchless mouse designed to enable remote control and enhance assistive technology for users with disabilities or limited mobility.

Background of the Invention

[0002] The proliferation of digital devices has led to an increasing reliance on input devices that depend on physical contact, such as keyboards, mice, and touchscreens. While these devices have served users for decades, they come with inherent limitations. Physical contact requires users to be in close proximity to the device, which can hinder ergonomic comfort and limit the potential for hands-free operation. Additionally, these existing input methods often struggle to provide an intuitive user experience, making them less suitable for a world where multitasking and mobility are paramount.

Prior Art

Many existing systems rely on rudimentary motion detection that can be hindered by environmental factors such as lighting conditions, background noise, and the complexity of human gestures. This results in a frustrating user experience characterized by lag, misinterpretation of gestures, and a lack of responsiveness. Similarly, European Patent No. EP3985491 B1 discloses a control method for touchless gesture control specifically focusing on user interaction with electronic displays. This method implements the detection of a user's finger in a virtual active area and a virtual hovering area to generate click or swipe events, with the help of depth sensors (e.g., like Gestoos and Leap Motion technologies). The motive is to provide an intuitive and hygienic way to interact with digital interfaces, reducing physical contact. While this patent offers a foundational approach to touchless interaction, its primary focus is on direct display manipulation, not the broader control of diverse hardware components for assisted living or robust remote operation in harsh industrial settings. US Patent No. US20130336524A1 implemented a design patent for a dynamic hand gesture recognition system that senses depth data for hand movement. This system functions by detecting a hand, extracting feature values from the depth data, and subsequently recognizing various

hand movements as specific gestures. This is achieved with the help of depth-sensing cameras (e.g., Kinect-like technology). The motive was to enable more intuitive and natural human-computer interaction, particularly for gaming and general computing, by allowing users to control devices without physical contact. While highly relevant to the core gesture recognition technology, this patent generally focuses on entertainment or general computing applications and does not explicitly address the specific challenges of low-latency, high-reliability control required for critical hardware operation by paralyzed individuals or in extremely hazardous industrial environments. US Patent No. US20080256494A1 implemented a design patent for a touchless hand gesture device controller that describes a system and method for controlling a device using free-space hand gestures. This patent details the use of a proximity sensor array to capture the hand's position and movement, which is then translated into control signals for a connected device. The motive was to provide a general-purpose non-contact input device that could replace or augment traditional mice and keyboards, enabling more natural and sanitary interaction. This prior art establishes a broad concept of touchless control; however, it does not delve into the specific nuances of adaptive control for varying levels of user mobility (e.g., for paralyzed individuals) or the specialized requirements for remote control in environments with extreme conditions, such as fluid ingress, limited visibility, or robust electromagnetic interference often found in industrial cleaning scenarios. US Patent No. US20210349537A1 developed a patent for remote device control via a virtual interface, enabling users to operate target devices (e.g., industrial manipulators, drones) through a VR/AR environment. This system uses a local computing device and interaction devices (e.g., smart rings) to receive user input in a virtual interface, which then generates and transmits control signals to the wirelessly connected target device. The motive is to eliminate the need for physical controllers and direct presence, making remote control more effective. This is relevant for operating hardware by paralyzed patients and machinery in hazardous environments, offering a framework for touchless remote control. Patent No. WO2017033359A1 designed a remote-controlled robot system primarily for manipulating robot arms to perform prescribed work. This system integrates multiple cameras for imaging robot arm operations, displaying transmitted captured images on a monitor, and allowing an operator to remotely manipulate the robot arm. While not explicitly "gesture-controlled" as a primary interface, the underlying

motive is to enable remote operation of machinery, particularly in industrial settings where direct human presence might be unsafe or impractical. This patent establishes the need and existing solutions for remote control in potentially hazardous environments, but its focus is on traditional joysticks or hand-grip manipulation for industrial robots, not the more intuitive and touchless gesture control using vision systems, which is crucial for reducing contamination risk and improving user interface for cleaning staff.

To address these challenges, there is a pressing need for advanced machine learning algorithms specifically designed for gesture recognition. These advanced systems would leverage the capabilities of machine learning to improve the accuracy and efficiency of gesture detection, allowing for a more natural and fluid interaction between users and devices. By utilizing design-driven approaches, such systems could learn from a user's unique gestures over time, adapting to their specific movements and preferences. This would not only enhance the user experience but also pave the way for hands-free and gesture-based controls that can be employed across various applications, from gaming to virtual reality and beyond.

The shortcomings of the existing systems have become increasingly critical as users demand more sophisticated and seamless methods of interacting with their devices, particularly in applications requiring quick responses and high accuracy. In conclusion, the evolution of input devices necessitates a shift towards advanced gesture recognition systems that prioritize real-time precision and user adaptability. By harnessing the power of machine learning, such innovations promise to create a seamless interaction experience, ultimately redefining how users engage with technology in a more intuitive and efficient manner. This invention aims to fill the existing gaps in current input methodologies, setting the stage for a new era of user-device interaction.

Object of the invention

[0003] The principal object of the invention is to enable hands-free interaction with digital systems, enhancing user convenience and accessibility. This is achieved through innovative technology that allows for intuitive control without the need for physical contact.

[0004] Another object of the invention is to enhance efficiency and safety in the cleaning and medical industry by enabling remote operation of cleaning devices

and medical equipment, thereby reducing the risk of contamination and exposure to pathogens. This advancement aims to improve operational workflows while maintaining high hygiene standards.

[0005] Another object of the invention is to provide a novel rehabilitation device that enables paralyzed patients to regain mobility and enhance their physical therapy experience. This device aims to improve the overall quality of life for individuals with mobility impairments.

Summary of the invention

[0006] This invention introduces a touchless mouse that operates through a real-time sensor integrated with a wearable sensor band. This innovative device leverages advanced sensing technology to detect hand movements and gestures in three-dimensional space, translating them into cursor movements and commands on a digital interface. The wearable sensor band is designed ergonomically for comfort and can be worn on the wrist, ensuring user convenience during operation. The system is equipped with a high-resolution motion sensor that captures subtle hand movements, allowing for precise control without physical contact with traditional mouse interfaces.

This invention presents an AI-driven recognition algorithm that processes the data collected by the wearable sensor band to generate control commands for digital systems. The algorithm utilizes machine learning techniques to continuously adapt and improve its accuracy based on user interactions, enabling a personalized experience tailored to individual usage patterns. This dynamic recognition capability facilitates seamless interaction with hardware, removing barriers associated with conventional input devices.

Furthermore, this invention is specifically designed to support assistive technology applications, making it particularly beneficial for individuals with mobility impairments or other disabilities. The touchless interface allows these users to interact with digital systems in a manner that is both intuitive and accessible, enhancing their ability to engage with technology effectively. The system's flexibility also makes it suitable for various environments, from home use to professional settings, ensuring that users can navigate digital platforms effortlessly and efficiently. Overall, this invention represents a significant advancement in user interface technology, promoting a more inclusive and user-friendly digital

experience.

Brief Description of Drawings

[0007] The foregoing and other features of embodiments will become more apparent from the following detailed description of embodiments when read in conjunction with the accompanying drawings. In the drawings, like reference numerals refer to like elements.

[0008] In the following description, for the purposes of explanation, numerous specific details are set forth to provide a thorough understanding of the embodiments of the invention. It is apparent, however, to one skilled in the art that the embodiments of the invention may be practiced without these specific details or with an equivalent arrangement. In other instances, well-known structures and devices are shown in block diagram form to avoid unnecessarily obscuring the embodiments of the invention.

FIG. 1 Illustrates a comprehensive system architecture diagram representing the components used in the invention, focusing on the embodiment of A method for operating a touchless mouse using gesture recognition to enhance user interaction and improve accessibility.

[0009] FIG.2 Illustrates a data flow diagram representing the logical information movement through the system in preparing the composition for A method for operating a touchless mouse using gesture recognition to enhance user interaction and improve accessibility.

[0010] FIG.3 Illustrates an activity diagram representing the operational flow and decision-making process scenario for A method for operating a touchless mouse using gesture recognition to enhance user interaction and improve accessibility as per one embodiment of the invention.

[0011] FIG.4 Illustrates a component interaction diagram representing the exemplary representation for A method for operating a touchless mouse using gesture recognition to enhance user interaction and improve accessibility as per one embodiment of the invention.

Detailed Description of the Invention

[0012] Reference will now be made in detail to the description of the present subject matter, one or more examples of which are shown in the figures. Each

example is provided to explain the subject matter and not a limitation. Various changes and modifications obvious to one skilled in the art to which the invention pertains are deemed to be within the spirit, scope and contemplation of the invention.

[0013] The integration of advanced gesture recognition technology with artificial intelligence enables a seamless and intuitive means of real-time control across various sectors. This innovative approach not only enhances operational efficiency but also significantly improves safety and accessibility, making it suitable for diverse applications. By allowing users to interact with systems through natural movements, it fosters precision in execution and reduces the risk of errors. As industries increasingly adopt this technology, the potential for transformative impacts grows, paving the way for smarter, more responsive environments that cater to the needs of users while ensuring a high standard of performance and safety.

[0014] The invention introduces an innovative approach to interaction by enabling touchless control through real-time finger gestures. This technology significantly enhances user experience by allowing seamless operation without physical contact, making it particularly beneficial in environments where hygiene and safety are paramount. Its versatility extends to various applications, particularly in hazardous medical and assistive scenarios, where traditional interfaces may pose risks or challenges. By harnessing the power of gesture recognition, users can engage with devices intuitively and efficiently, thereby improving accessibility and usability in critical settings. This advancement represents a meaningful step forward in human-device interaction technology.

[0015] The invention introduces a sophisticated approach to modern drainage systems by harnessing the power of advanced robotics intertwined with smart technology. This integration facilitates seamless communication between devices, enabling efficient monitoring and management of drainage processes. Moreover, the system supports instantaneous data relay, which proves invaluable for applications such as remote medical operations. By allowing real-time updates and insights, the enhancement not only optimizes operational efficiency but also ensures that critical information is readily available for timely decision-making in various scenarios. This fusion of robotics and smart technology marks a significant advancement in both infrastructure and healthcare domains.

[0016] Technology significantly enhances operational efficiency by streamlining processes that typically require human involvement in hazardous settings. By leveraging automation, it not only minimizes risks but also alleviates the financial burden associated with traditional infrastructure and travel, particularly in remote medical scenarios. This innovative approach presents an economical alternative for those with physical challenges, ensuring that essential services are accessible without imposing excessive costs. The integration of this technology fosters a more sustainable model, ultimately benefiting users by providing reliable solutions that enhance independence while promoting safety and reducing expenses across various applications.

[0017] FIG. 1. The system architecture represents a layered, end-to-end design for a gesture- and voice-controlled interaction system, clearly separating sensing, intelligence, execution, and feedback to ensure reliability and scalability. The architecture begins with the User, who initiates interaction using physical hand or body gestures or by issuing a voice command. These physical actions are captured by 101: Camera / Sensor, which acts as the primary perception layer and may include cameras, motion sensors, depth sensors, or wearable sensors depending on deployment. The raw image frames or sensor signals produced at this stage are inherently noisy and unstructured, which is why they are forwarded to 103: Pre-Processing, Noise Reduction & Segmentation. This module removes environmental noise, filters irrelevant background data, and segments the gesture region so that only meaningful motion information is preserved. The refined output is passed to 105: Feature Extraction, where the system converts visual or motion patterns into numerical feature vectors such as hand trajectories, joint angles, motion velocity, or spatial relationships. These feature vectors form the input to 107: Gesture Recognition ML / DL Models, which represent the intelligence layer of the architecture and use trained machine learning or deep learning models to classify the gesture and generate a unique Recognized Gesture ID. This recognized intent is forwarded to 109: Command Generation, which maps abstract gesture identifiers to concrete system-level actions, translating human intent into executable control commands. These commands are then executed by 111: Operating System / Device, which serves as the control and execution layer, interfacing directly with the target hardware, embedded system, or application environment. Once execution is complete, the system ensures closed-loop

interaction by generating user feedback through two parallel channels: 113: Visual / Audio Feedback, which may include screen notifications, indicators, or alerts, and 115: Voice Output Module, which produces spoken confirmations or system responses. Together, these components form a cohesive architecture that transforms raw human gestures into intelligent, validated actions with real-time feedback, making the system intuitive, responsive, and suitable for real-world human–machine interaction.

[0018] FIG. 2. Data Flow Diagram (DFD) explains how information moves through the system from the moment the system is activated until an action is completed and stored, focusing purely on data transformation rather than physical components. The flow begins at 201: System Activation Signal, which represents the initial trigger that powers up the system or wakes it from an idle state. Once activated, the system enables 203: Wearable, such as a smart glove or band, which captures raw motion or touch input generated by the user's gesture. This unprocessed movement data is forwarded as raw motion or touch data to 205: Sensor, where the signals are continuously collected and streamed. The resulting sensor signal stream is then sent to 207: Gesture Input, a processing stage that structures the incoming data into a consistent and machine-readable format. This structured gesture data flows into 213: Gesture Recognition, where recognition algorithms analyze the motion patterns and determine the most likely gesture performed by the user. Along with producing recognized gesture information, this stage also generates recognized status and logs, ensuring that system decisions are traceable and can be monitored for accuracy. The recognized gesture data is stored in 215: Cloud Record, allowing historical tracking, learning, or system analytics. Simultaneously, the recognized gesture information is passed to 209: Command Generation, which translates human intent into a precise control instruction packet that the system can execute. This packet is then delivered to 211: Command Execution, where the actual action is carried out on the target hardware or software system. Finally, the executed action and its corresponding gesture details are stored as stored gesture and action data, leading to 217: End, which marks the completion of one full interaction cycle. This data flow diagram clearly shows how raw human input gradually becomes structured data, recognized intent, executable commands, and stored system knowledge.

[0019] FIG. 3 The activity diagram describes how the system behaves internally

from the moment a user performs a gesture until the system completes an action and waits for the next interaction, focusing on actions and decision points rather than data or components. The flow begins with 301: User Gesture, where the user initiates interaction by performing a physical hand or body movement. This gesture is captured during 303: Gesture Acquisition, which represents the continuous sensing of motion through cameras or sensors and results in raw gesture data. Since raw data is often noisy and inconsistent, the system moves to 305: Preprocessing, Noise Reduction and Segmentation, where background interference is removed and the actual gesture region is isolated. The cleaned data then proceeds to 309: Extract Gesture Features, where meaningful characteristics such as motion direction, speed, and spatial patterns are converted into structured feature vectors. At 311: Gesture Detected? The system checks whether a valid gesture is present; if no meaningful pattern is detected, the flow safely returns to 325: Await Next Gesture, ensuring the system does not execute unintended actions. If a gesture is detected, the process continues to 313: Classify and Recognize Gesture, where machine learning or pattern-based logic identifies the gesture type. The system then evaluates reliability at 315: Recognition Confidence Valid? confirming whether the recognition meets accuracy thresholds; if confidence is low, the system again returns to the waiting state to avoid incorrect execution. When confidence is sufficient, 317: Generate Control Command translates the recognized gesture into a system-understandable instruction. This instruction is executed in 319: Execute Command via OS, where the operating system or control layer performs the intended action. Finally, the system responds to the user through 321: Generate Visual Feedback and 323: Generate Voice Feedback, confirming successful execution before returning to 325: Await Next Gesture. This activity diagram clearly illustrates safe decision handling, validation, and continuous interaction flow.

[0020] FIG.4. The component interaction diagram illustrates how the major system modules cooperate with each other during real-time operation, emphasizing modular design and clear responsibility boundaries between components. The interaction begins with 401: User, who initiates communication with the system either through physical gestures or spoken voice commands. These inputs are captured by 403: Wearable / Input Interface, which may include smart gloves, motion sensors, or embedded input devices that translate human actions into digital

signals. In parallel, voice-based interaction is handled by 405: Voice Input Module, which captures spoken commands and converts them into a digitized audio signal suitable for processing. Both gestures and voice data are transmitted through 407: Wireless Communication Module, highlighting the system's ability to operate in a distributed or remote environment while maintaining low-latency communication. The wireless module ensures that all input data reaches 409: Target Hardware / System, which represents the core processing and execution unit responsible for interpreting commands and performing the requested actions. Once the target system processes the instruction and completes execution, it generates response data that flows back into the feedback components. Visual execution status and system responses are directed to 411: Visual Feedback Module, which may display confirmations, alerts, or system states on a screen or interface. Simultaneously, auditory responses are routed to 413: Audio / Voice Feedback Module, which converts system responses into spoken feedback for the user. Both feedback channels ultimately communicate back to the User, ensuring clear acknowledgment of system actions and maintaining a closed interaction loop. This component interaction diagram clearly demonstrates how independent modules interact without tight coupling, allowing the system to remain scalable, adaptable, and easy to maintain while supporting multimodal human–computer interaction.

[0021] The invention revolutionizes interaction by enabling gesture-based control, which eliminates the need for physical touch, thus enhancing hygiene and accessibility. It innovatively adapts to user-specific needs, catering to medical operators, cleaners, and individuals with disabilities, including those who are paralyzed. This adaptability not only improves the overall user experience but also enhances system responsiveness, allowing for seamless interaction. Additionally, the invention incorporates advanced learning algorithms that improve gesture recognition over time, ensuring that the system becomes more intuitive and efficient in recognizing and responding to user commands, ultimately setting it apart from existing technologies.

[0022] The inventive step is the development of a system that enables remote surgeries with precise gesture control, facilitating long-distance operation of medical devices while ensuring hygiene by offering a touchless interface.

[0023] The touchless finger mouse utilizes real-time gesture recognition and

wireless communication to facilitate seamless interaction with digital systems. By employing AI-driven models, it achieves high accuracy in recognizing gestures and converting them into actionable commands. This innovation proves invaluable in hazardous environments, such as drainage cleaning, where operators can remotely control IoT-enabled robots, thereby minimizing direct human exposure to danger. In the medical field, it enables doctors to conduct surgeries from remote locations by skillfully guiding robotic instruments through hand gestures. Furthermore, the system offers significant advantages for paralyzed patients, allowing them to communicate and control various devices using simple finger movements. Its touchless functionality enhances hygiene, making it particularly suitable for critical environments like operating rooms. The lightweight and wearable design of the device ensures user convenience and ease of use, making it a versatile tool across multiple applications. Overall, the touchless finger mouse stands out as a transformative solution for enhancing human interaction with technology.

[0024] The invention offers several advantages, including an advanced gesture recognition algorithm; dynamic context-aware control; real-time feedback; and enhanced usability in sterile environments.

[0025] The invention achieves its advantages through a sophisticated remote control system that allows users to operate devices from a safe distance, significantly mitigating risks in hazardous environments. For paralyzed patients, the system utilizes gesture recognition technology, enabling them to interact with devices effortlessly, thus enhancing their independence and quality of life. Its precision in control is particularly beneficial in medical applications, where accurate maneuvering is critical for patient safety and treatment effectiveness. The touchless interface promotes hygiene by eliminating physical contact, which is essential in maintaining sterility in medical settings. Additionally, the invention is designed for cost-effective integration with a variety of devices, ensuring that it remains accessible and practical for widespread use without imposing significant financial burdens.

CLAIMS:

We Claim,

1. A system for operating a touchless mouse using finger gesture recognition, comprising: a plurality of wearable sensor bands equipped with high-resolution motion sensors; an AI-driven recognition algorithm; to obtain hand movement data from the user's gestures; determine cursor movements in three-dimensional space by the at least one motion sensor based on the captured gestures; process the data to generate control commands; control digital interface actions using the recognition algorithm based on the determined cursor movements.
2. The system of claim 1, wherein the wearable sensor bands include ergonomic features designed for user comfort.
3. The system of claim 1, wherein the cursor movements are calculated using a machine learning model that adapts based on user interaction patterns.
4. The system of claim 1, wherein the additional criterion includes real-time responsiveness to hand gestures for immediate cursor control.
5. The system of claim 4, wherein the real-time responsiveness is defined by a latency of less than 50 milliseconds in translating gestures to cursor movements.

Dated: 09/03/2026

"A Method for Operating a Touchless Mouse Using Gesture Recognition to Enhance User Interaction and Improve Accessibility"

ABSTRACT

This invention relates to a method for operating a touchless mouse using finger gesture recognition to enhance user interaction and improve accessibility. The system comprises a wearable finger band equipped with a high-resolution motion sensor that captures hand movements and gestures in three-dimensional space, translating them into cursor movements and commands on a digital interface. An AI-driven recognition algorithm processes sensor data, generating control commands for various digital systems, including IoT devices and medical equipment. This dynamic and adaptive algorithm leverages machine learning to tailor the user experience based on individual interaction patterns. Designed ergonomically for comfort, the finger band facilitates intuitive interaction, particularly benefiting individuals with mobility impairments. By eliminating the need for physical contact with traditional input devices, this invention promotes a more inclusive and user-friendly digital experience across diverse environments, from home to professional settings, thereby enhancing overall accessibility in technology.

Dated: 09/03/2026

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[On behalf of all Applicants, Refer form 26]

To,
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